Linear inequalities

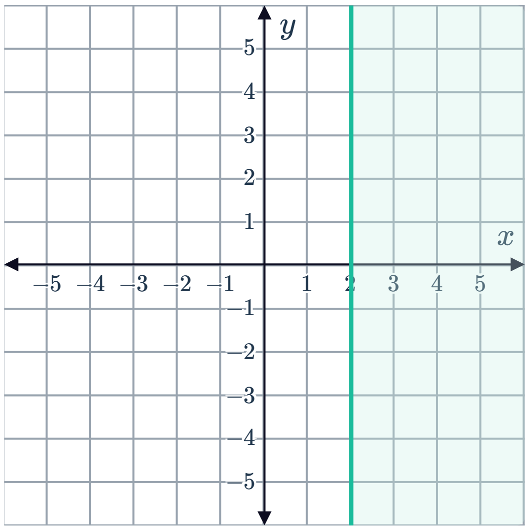
1 State whether the following inequalities are represented using a solid line or a dashed line on a graph:

a $y - 3x > 5$ **b** $x \geq 4$ **c** $-2y + x \leq -10$ **d** $y + 3x > 7$

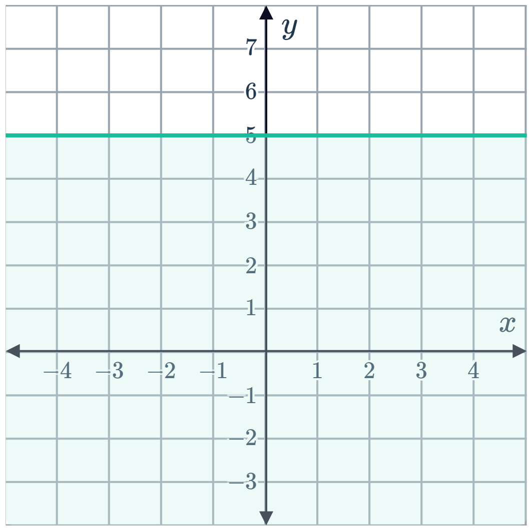
e $-y - 6x > 0$ **f** $y + 3x \leq -10$ **g** $y \geq -2$ **h** $3y + 3x > 4$

2 Write the inequality that describes the following shaded regions:

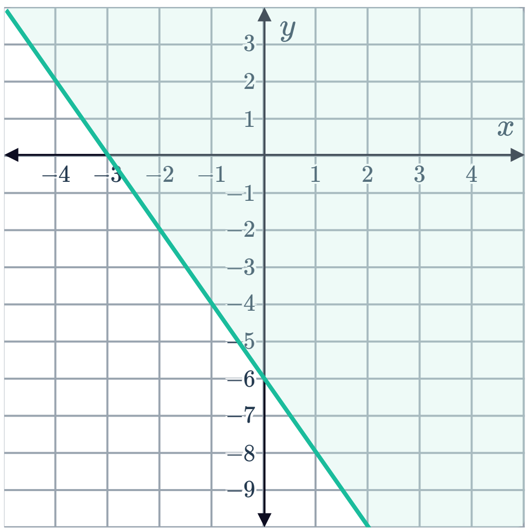
a



b



c



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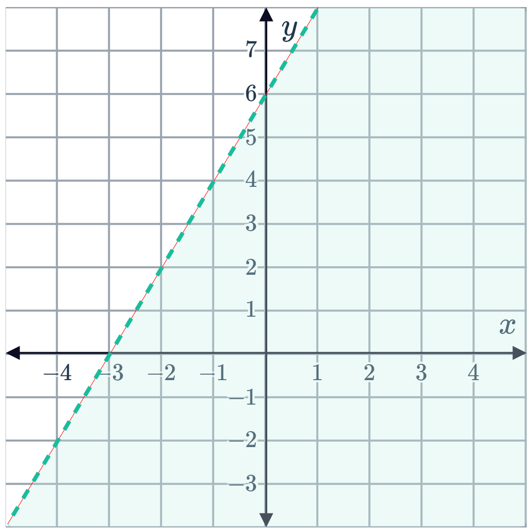
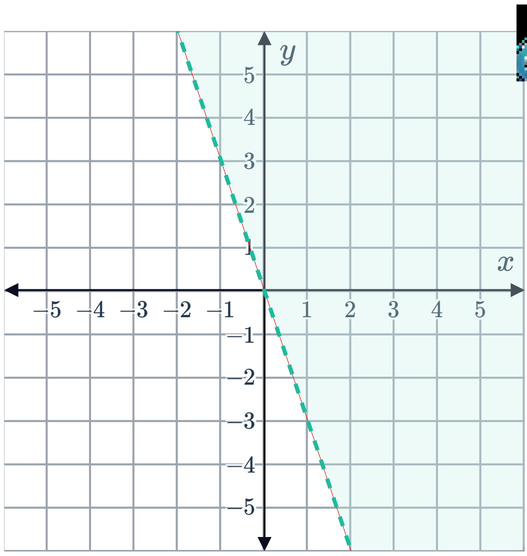
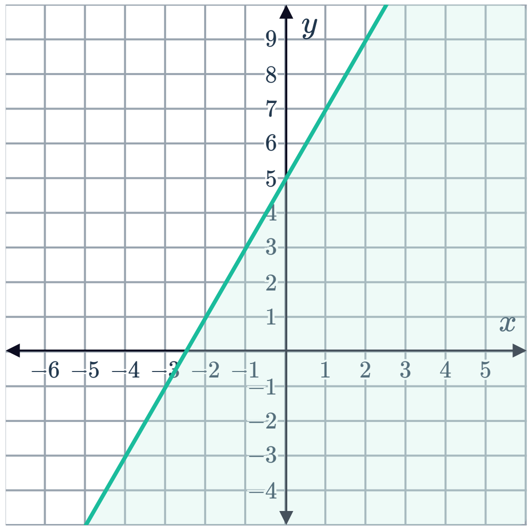


e



f



3 Is $(6,20)$ a solution of $y \geq 4x+5$?

4 Is $(3,2)$ a solution of $3x+2y \geq 12$?

5 Determine which point satisfies each of the following inequalities:

a $x-y>7$

A $(-1,-7)$ **B** $(8,-5)$ **C** $(13,6)$ **D** $(0,0)$

b $y>2x$

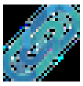
A $(9,15)$ **B** $(-1,-2)$ **C** $(1,4)$ **D** $(0,0)$

c $y>-10x-2$

A $(5,-52)$ **B** $(-1,3)$ **C** $(-3,26)$ **D** $(1,-11)$

Graphing linear inequalities

6 Describe the three steps required to graph the inequality $y+7x \geq 5$.

7 Consider the inequality $y < -3x - 6$. 

a Solve for the x and y-intercepts of the line $y = -3x - 6$.

b Determine which of the following points satisfy the inequality $y < -3x - 6$:

A $(-7, 2)$ **B** $(-6, 3)$ **C** $(3, -2)$ **D** $(-6, -3)$

c Hence sketch the graph of $y < -3x - 6$.

d Do the points on the line satisfy the inequality?

8 Consider the inequality $y \leq -2x + 2$.

a Solve for the x and y intercepts of the line $y = -2x + 2$.

b Determine which of the following points satisfies the inequality $y \leq -2x + 2$:

A $(2, 3)$ **B** $(3, -6)$ **C** $(4, -2)$ **D** $(1, 2)$

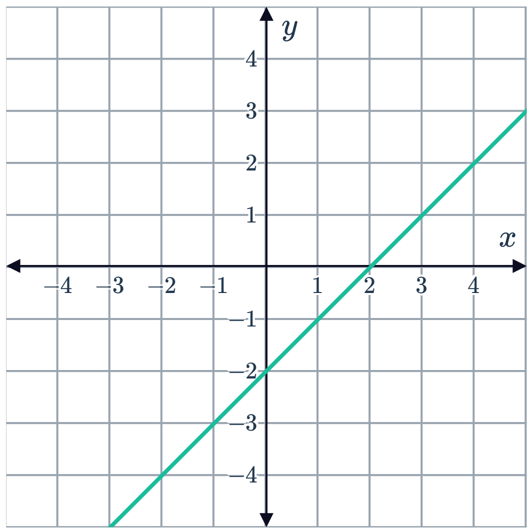
c Hence sketch the graph of $y \leq -2x + 2$.

d Do the points on the line satisfy the inequality?

9 The solutions to the equation $y = x - 2$ are plotted on the following graph:

a Graph the solutions to $y > x - 2$.

b Graph the solutions to $y \leq x - 2$.



10 Sketch the graphs of the following inequalities:

a $y \leq x - 2$ **b** $5x + 2y \geq 20$ **c** $4x + 3y > 12$ **d** $6x + 3y > 12$

e $y > 2x - 4$ **f** $y \leq 3x - 4$ **g** $y < 3x - 9$ **h** $y \geq -2x + 4$

i $2x + 3y < 12$ **j** $y > 2x - 4$ **k** $y \leq x + 5$

11 State the linear inequality that satisfies the following:

a A linear inequality has solutions of $(0, -5)$ and $(-1, -9)$ on its boundary line and is not satisifed by $(1, 3)$.

b A linear inequality has solutions of $(0, -6)$ and $(-6, -24)$ on its boundary line and is satisifed by $(-3, -10)$.

Applications

12 Luigi has several bottles of lemonade and Pepsi that he wants to take to a picnic, but he only has enough room in his backpack for 9 bottles.

Let x represent the number of bottles of lemonade that he takes to the picnic.

Let y represent the number of bottles of Pepsi that he takes to the picnic.

a Write an inequality to represent the situation.

b Graph the inequality on a cartesian plane.

c Use the graph to determine whether Luigi can take the following combinations of bottles to the picnic:

i 1 bottle of lemonade and 11 bottles of Pepsi

ii 3 bottles of lemonade and 11 bottles of Pepsi

iii 3 bottles of lemonade and 4 bottles of Pepsi

iv 5 bottles of lemonade and 3 bottles of Pepsi

13 Airline passengers are told to be at the gate no less than 15 minutes before a flight's departure. It takes Gwen 30 minutes to get to the airport, and y minutes to go through check in and security, depending on how busy the airport is.

a Form an inequality relating x and y, where x represents the number of minutes left until Gwen's flight departs.

b Graph the inequality on a cartesian plane.

c If the check-in and security process takes 20 minutes, how many minutes before her flight's departure can Gwen leave her house at the latest?

14 Roxanne has set aside $\$14.00$ in her shopping budget for fruit this week. Currently, oranges are on sale for $\$1.40$ each, while apples are on sale for $\$1.75$ each.

Let x represent the number of oranges that Roxanne buys.

Let y represent the number of apples that Roxanne buys.

a Write an inequality to represent the situation.

b Graph the inequality on a cartesian plane.

c Use your graph to determine whether Roxanne can buy the following combination of fruits:

i 6 oranges and 7 apples

ii 3 oranges and 4 apples

iii 7 oranges and 4 apples

iv 8 oranges and 1 apples